

ON THE LIE SOLVABILITY OF NOVIKOV ALGEBRA

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Abstract

This project was completed under the guidance of Prof. Saber Ahmed¹. My aim was to learn about Lie Algebras and Lie Solvability from [1] and use it to understand the theorem from the paper [2] which states that any Novikov algebra over a field of characteristic $\neq 2$ is Lie-solvable iff its commutator ideal $[N, N]$ is right nilpotent.

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1 Introduction

Lie Algebra has its roots in the works of Marius Sophus Lie who studied transformations groups called Lie Groups. This algebra was introduced around 1870s to study the concept of infinitesimal transformations. The name *Lie Algebra* was coined by Herman Weyl in 1930s. In recent history, both Lie groups and Lie algebra have become an essential part of mathematics and theoretical physics. Lie groups are groups which are smooth manifolds and

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any Lie group gives rise to a Lie algebra which is its tangent space at the identity (maybe thought of as infinitesimal symmetry of motions). Conversely, to any finite-dimensional Lie algebra over real or complex numbers, there is a corresponding connected Lie group unique up to finite covering. This correspondence allows one to study structure and classification of Lie groups in terms of their Lie algebras. Further, we also have Lie superalgebras which can be explained as graded (decomposition of vector space into a direct sum of vector spaces) Lie algebras which are used to study the behaviour of quantum particles. However, my motivation to do this project lies in the fact that Lie algebras and their representations are a central object of interest in their own right.

2 Introduction to Lie Algebra

Definition 2.1. Let F be a field. A Lie Algebra is an F -vector space L together with a bilinear map, also called the Lie bracket,

$$L \times L \rightarrow L, \quad (x, y) \mapsto [x, y] \quad (1)$$

satisfying the following properties:

- (a) $[x, x] = 0$ for all $x \in L$
- (b) $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$ for all $x, y, z \in L$.

The Lie bracket $[-, -]$ is also called the *commutator* of x and y . The second property is known as the *Jacobi Identity*. Observe that, from the bilinearity of the bracket we have:

$$0 = [x + y, x + y] = [x, x] + [x, y] + [y, x] + [y, y] = [x, y] + [y, x]. \quad (2)$$

Thus, we can conclude $[x, y] = -[y, x]$ (skew-symmetry) for all $x, y \in L$. If the field F does not have characteristic 2 (i.e symmetry = skew-symmetry), then putting $x = y$ in (2) gives us property (a).

2.1 Examples

1. Let $F = \mathbf{R}$. The wedge product $(x, y) \mapsto x \wedge y$ defines the structure of a Lie

algebra on $\mathbf{R}^3$. If $x = (x_1, x_2, x_3)$ and $y = (y_1, y_2, y_3)$, $x \wedge y = \begin{vmatrix} i & j & k \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix} = (x_2y_3 - x_3y_2, x_1y_3 - x_3y_1, x_1y_2 - x_2y_1)$.

2. The trivial Lie algebra, i.e., given any vector space V , the lie bracket is defined by $[x, y] = 0 \quad \forall x, y \in V$. This implies that any two elements of this algebra commute. This is known as an *Abelian Lie Algebra*. The field F may be considered as a one - dimensional Lie algebra.
3. Now we define an important example which is the set of all linear endomorphisms $gl(V) := \{\phi \mid \phi \text{ is an endomorphism from } V \rightarrow V\}$. This is a vector space over F . It becomes a Lie algebra called the *general linear algebra* under the bracket:

$$[x, y] := x \circ y - y \circ x,$$

where $\circ$ denotes function composition. We can also represent the linear maps as matrices by fixing a basis over the space $gl(n, F)$ which is the vector space of all $n \times n$ matrices over the field F with the bracket $[x, y] = xy - yx$. The basis of $gl(n, F)$ consists of matrix units e_{ij} for $1 \leq i, j \leq n$. Here, e_{ij} represents an $n \times n$ matrix which has entry 1 in the ij -th position and entry 0 in every other position. Note that, $[e_{ij}, e_{kl}] = \delta_{jk}e_{il} - \delta_{il}e_{kj}$ where δ_{ij} is the Kronecker delta.

4. We end our list by specifying another important class of matrices, which are the *special linear matrices* defined as

$$sl(n, F) = \{A \in gl(n, F) \mid \sum_{i=j} a_{ij} = 0\}.$$

For arbitrary square matrices, X and Y , $XY - YX$ has trace zero, hence, $[x, y] := xy - yx$ defines a Lie algebra on $sl(n, F)$. As a vector space, the basis of $sl(n, F)$ are e_{ij} with $i \neq j$ and $h_i = e_{ii} - e_{i+1, i+1}$. Note that $sl(n, F) \subseteq gl(n, F)$.

2.2 Subalgebra & Ideals

Definition 2.2. Given a Lie algebra L , we can define a Lie subalgebra of L to be a vector space $K \subseteq L$ such that $[x, y] \in K$, $\forall x, y \in K$.

Definition 2.3. An ideal of a Lie algebra L is a subspace I of L such that $[x, y] \in I \forall x \in L, y \in I$.

Observe that, since $[x, y] = -[y, x]$, we do not need to distinguish between left and right ideals. Also, an ideal is always a subalgebra but a subalgebra need not be an ideal. The subalgebra $Z(L) := \{x \in L \mid [x, y] = 0 \forall y \in L\}$ called the center of L is an ideal of L . It consists of all the elements in L that commute with every other elements of L .

2.3 Homomorphisms

Definition 2.4. If L_1, L_2 are Lie algebras over a field F , we say that a map $\phi : L_1 \rightarrow L_2$ is a homomorphism if:

- ϕ is a linear map
- $\phi([x, y]) = [\phi(x), \phi(y)] \forall x, y \in L_1$.

Note that the $[-, -]$ in RHS is the Lie bracket in L_2 while the bracket in LHS is in L_1 . ϕ is an isomorphism if it is also bijective.

Example: The adjoint homomorphism, defined as $ad : L \rightarrow gl(L)$ by $ad_x(y) := [x, y]$. **Remark:** The map ad_x can be understood as the adjoint of a linear map which is the transpose of a matrix w.r.t dual bases.

To show that the map ad_x is a homomorphism, we first show that it is linear which is obvious from the definition. To check conservation of bracket operation,

$$ad_z([x, y]) = [z, [x, y]] = [x, [z, y]] + [y, [z, x]] = [ad_z(y), ad_z(x)],$$

hence, we prove that the adjoint map is an homomorphism.

2.4 Derivations

Let A be an algebra over a field F . A *derivation* of A is a F -linear map, $D : A \rightarrow A$ such that $D(ab) = aD(b) + D(a)b \forall a, b \in A$. We denote the set of derivations of A by $Der A$. $Der A$ is a vector subspace of $gl(A)$ since it is closed under addition, scalar product and contains the zero linear map.

$Der A$ is a Lie subalgebra of $gl(A)$ under the following bracket: $[D, E] := D \circ E - E \circ D$ which is also a derivation, since it is a subspace of $gl(A)$.

Example: Let $A = C^\infty(\mathbf{R})$ be the vector space of infinitely differentiable functions over $\mathbf{R}$. For $f, g \in A$, we define fg as pointwise multiplication. The usual derivative, $D(f) = f'$ is a derivation of A since,

$$D(fg) = (fg)' = f'g + fD(g) = D(f)g + fD(g).$$

2.5 Construction of Ideals

In this section we understand how we can construct more Lie algebras from the existing ones. If I and J are ideals of a Lie algebra then $I \cap J$ is also an ideal of L . We know that $I \cap J$ is a subspace of L , so, we need to check if $x \in L$ and $y \in I \cap J$ then $[x, y] \in I \cap J$, which is true since $[x, y] \in I$ and $[x, y] \in J$ since, I & J are both ideals. **Example:**

- (a) $I + J := \{x + y \mid x \in I, y \in J\}$ where I and J are ideals, is an ideal of L .
- (b) $[I, J] := \text{span}\{[x, y] \mid x \in I, y \in J\}$ which is the span of all commutators is also an ideal of L . We can easily verify this since, for $x \in I, y \in J, u \in L$, the Jacobi identity gives us $[u, [x, y]] = [x, [u, y]] + [[u, x], y] \in [I, J]$. When $I = J = L$, the ideal $[L, L]$ is called the derived algebra of L and is denoted by L' . Observe that it is necessary to consider the linear span since the set of commutators may not be a subspace; the sum of two commutators is not guaranteed to be a commutator. Let $t = \sum c_i [x_i, y_i]$, where the c_i are scalars and $x_i \in I$ and $y_i \in J$. Then, for any $u \in L$, we have

$$[u, t] = [u, \sum c_i [x_i, y_i]] = \sum c_i [u, [x_i, y_i]], \quad (\text{bilinearity})$$

where $[u, [x_i, y_i]] \in [I, J] \implies [u, t] \in [I, J]$, hence, $[I, J]$ is an ideal of L .

- (c) If I is an ideal of the Lie algebra L , then I is in particular a subspace of L , and so we can consider the cosets $z + I := \{z + x \mid x \in I\}$ for $z \in L$ and the quotient vector space $L/I = \{z + I \mid z \in L\}$. We can define a Lie bracket on L/I as follows: $[w + I, z + I] := [w, z] + I$. This is called a *Quotient Algebra*.

We need to check whether this bracket is well - defined. Suppose

$$w + I = w' + I, z + I = z' + I$$

$$\implies w - w' \in I, z - z' \in I$$

$$[w', z']$$

$$= [w' + (w - w'), z' + (z - z')]$$

$$= [w, z] + [w - w', z] + [w', z - z'] + [w - w', z - z']$$

$$= [w, z] + I = [w, z].$$

3 Solvability of Lie Algebra

Consider the ideal I of a Lie algebra L . We ponder on when the quotient algebra L/I is abelian.

Lemma 3.1. *Suppose that I is an ideal of L . Then L/I is abelian if and only if I contains the derived algebra L' .*

Proof. The algebra L/I is abelian iff $\forall x, y \in L$ we have,

$$[x + I, y + I] = [x, y] + I = I,$$

or equivalently for all $x, y \in L$ we have $[x, y] \in I$. Since I is a subspace of $L \Leftrightarrow \text{span}\{[x, y]\} \subseteq I$ (since for all $x, y \in L$, $[x, y] \in I$). Hence, $L' \subseteq I$. $\square$

So, suppose we have an ideal $J \subseteq L'$ such that L/J is abelian $\implies J = L'$. Hence, this lemma tells us that the derived algebra L' is the smallest ideal of L with an abelian quotient. Now, by a similar argument the derived algebra has a smallest ideal whose quotient is abelian which is the derived algebra of L' , which is denoted as $L^{(2)}$. We can define a *derived series* of L to be the series with terms:

$$L^{(1)} = L', \quad L^{(k)} = [L^{(k-1)}, L^{(k-1)}],$$

for $k \geq 2$. Then, $L \supseteq L^{(1)} \supseteq L^{(2)} \supseteq \dots$. Observe that, $L^{(k)}$ is an ideal of L and not just of $L^{(k-1)}$, since product of ideals is also an ideal.

Definition 3.1. *The Lie algebra L is said to be **solvable** if for some $m \geq 1$, we have $L^{(m)} = 0$.*

Example:

1. Assume that $L^{(1)}$ is 1 - dimensional, then $L^{(1)} \subseteq Z(L)$. The algebra with basis f, g, z such that $[f, g] = z \subseteq Z(L)$ is called the *Heisenberg Algebra*. Such an algebra exists since: Consider $f, g \in L$ such that $[f, g] \neq 0$. Since L' is 1 - dimensional, hence, $[f, g]$ spans L' . Also, $L' \subseteq Z(L) \implies [[f, g], x] = 0, x \in L$. Let, $z := [f, g]$. The linear independence of f, g, z can be verified by a simple check. the Heisenberg Algebra thus exists, and is solvable since, $L^{(2)} = 0$.
2. The algebra of upper triangular matrices, $b(n, F)$ is also solvable. If $L = b(n, F)$, we have $[x, y]$ strictly upper triangular $\forall x, y \in L$ (since $A \cdot B$ and $B \cdot A$ have same diagonal entries, so $A \cdot B - B \cdot A$ is strictly upper triangular). So, $\text{span}\{[x, y]\} = L' \subseteq \eta(n, F)$. Basis for $b(n, F) = \{e_{ij} \mid 1 \leq i \leq j \leq n\}$, whereas basis for $\eta(n, F)$ is $\{e_{ij} \mid 1 \leq i < j \leq n\}$. Note that in $b(n, F)$, $[e_{ii}, e_{ij}] = e_{ij}$ for $i < j$. Hence $L' = \eta(n, F)$. For $i < j$, we define the *level* of e_{ij} as the integer $j - i$. Then $L^{(2)}$ is spanned by the commutators of the form $[e_{ij}, e_{kl}]$ where $i < j$ and $k < l$. But observe that for $i < j$ and $k < l$

$$[e_{ij}, e_{kl}] = \begin{cases} -e_{kj} & \text{if } i = l \\ e_{il} & \text{if } j = k \\ 0 & \text{if otherwise,} \end{cases} \quad (3)$$

Notice that if $i = l$, then $j \neq k$ and the level of $e_{kj} = j - k = (j - i) + (l - k) \geq 2$. Similarly, level of $e_{il} \geq 2$. Hence, $L^{(2)}$ is spanned by e_{pq} which has level atleast 2. One can easily prove by induction on i that $L^{(i)}$ is spanned by e_{pq} with level atleast $2^{(i-1)}$. Since, level of any $e_{pq} \leq n - 1$, it is clear that $L^{(i)} = 0$ whenever $2^{(i-1)} \geq n - 1$. Hence, L is solvable.

Lemma 3.2. *If L is a Lie algebra with ideals*

$$L = I_0 \supseteq I_1 \supseteq I_2 \cdots \supseteq I_{m-1} \supseteq I_m = 0,$$

such that I_{k-1}/I_k is abelian for $1 \leq k \leq m$, then L is solvable.

Proof. We want to show $L^{(k)} \subseteq I_k$ for any $1 \leq k \leq m$.

Now L/I_1 is abelian $\implies L' \subseteq I_1$. Suppose $L^{(k-1)} \subseteq I_{k-1}$ (inductive hypothesis) where $k \geq 2$. We know, I_{k-1}/I_k is abelian $\implies [I_{k-1}, I_{k-1}] \subseteq I_k$. Hence, by hypothesis, $L^{(k)} = [L^{(k-1)}, L^{(k-1)}] \subseteq [I_{k-1}, I_{k-1}] \subseteq I_k$. Hence, by induction, $L^{(k)} \subseteq I_k$. Putting $k = m$, we are done. $\square$

Lemma 3.3. *Let L be a Lie algebra.*

- (a) *If L is solvable, then every subalgebra and every homomorphic image of L is solvable.*
- (b) *Suppose that L has an ideal I such that I and L/I are solvable. Then, L is solvable.*
- (c) *If I and J are solvable ideals of L , then $I + J$ is a solvable ideal of L .*

Proof. (a) From the definition, if K is a subalgebra of L , then $K^{(i)} \subseteq L^{(i)}$. Similarly, if $\phi : L \rightarrow M$ is an epimorphism, an easy induction on i shows that $\phi(L^{(i)}) = M^{(i)}$.

- (b) Say, $(L/I)^{(n)} = 0$. Applying part (a) to the canonical homomorphism, $\pi : L \rightarrow L/I$, we get $\pi(L^{(n)}) = 0$, or $L^{(n)} \subset I = \ker \pi$. Now, if $I^{(m)} = 0$, the obvious fact that $(L^{(i)})^{(j)} = L^{(i+j)} \implies L^{(n+m)} = 0$.
- (c) One of the standard homomorphism theorems yields an isomorphism between $(I+J)/J$ and $I/(I \cap J)$. As a homomorphic image of I , the right side is solvable, so $(I+J)/J$ is solvable. Then so is $I+J$ from part (b). $\square$

4 Nilpotency of Lie Algebra

We define the *lower central series* of a Lie algebra L to be the series with terms

$$L^1 = L' \quad L^k = [L, L^{k-1}] \quad \text{for } k \geq 2.$$

Then $L \supseteq L^1 \supseteq L^2 \supseteq \dots$. As the product of ideals is an ideal, L^k is even an ideal of L . Note that the quotient L^k/L^{k+1} is contained in the center of L/L^{k+1} , hence the name "central series".

Definition 4.1. *The Lie algebra L is said to be **nilpotent** if for some $m \geq 1$ we have $L^m = 0$.*

Lemma 4.1. *Any nilpotent Lie algebra is solvable.*

Remark: However, every solvable Lie algebra is not nilpotent; one example would be $b(n, F)$.

Proof. It can be easily seen by inducting on i to show $L^{(i)} \subseteq L^i$ □

Lemma 4.2. *Let L be a Lie algebra.*

- (a) *If L is nilpotent, then any Lie subalgebra of L is nilpotent.*
- (b) *If $L/Z(L)$ is nilpotent, then L is nilpotent.*

Proof. Part(a) is clear from definition. From induction we can show that $(L/Z(L))^k = (L^k + Z(L))/Z(L)$. So if $(L/Z(L))^m = 0$, then $L^m \subseteq Z(L) \implies L^{m+1} = 0$. □

5 Novikov Algebra

Definition 5.1. *An algebra N over a field K is called a Novikov Algebra if it satisfies the following:*

- (1) $(x, y, z) = (y, x, z)$
- (2) $(xy)z = (xz)y$,

where $(x, y, z) = (xy)z - x(yz)$ is called the associator of the elements x, y, z .

Example: Consider $K[x]$ to be the algebra of all polynomials in one variable x over a field K . Consider $K[x]$ as a differential algebra with derivation $\partial = \frac{\partial}{\partial x}$. Then $K[x]$ becomes a Novikov algebra w.r.t the product $f \circ g = fg'$. This algebra is also a well-known Witt algebra W_1 .

We say that a Novikov algebra N is solvable if the Lie algebra $N^{(-)}$ is solvable, where $N^{(-)}$ denotes the Lie algebra of N with respect to the commutator $[x, y] = xy - yx$. We say that N thus becomes *Lie admissible*. The authors in their paper [2] noticed that the space of commutators $[N, N]$ of a Novikov algebra N is an ideal of N over a field of characteristic $\neq 2$. Since it is an ideal, we can talk about the solvability and nilpotency of N , and they further prove that Novikov algebra N over a field of characteristic $\neq 2$ is Lie-solvable iff $[N, N]$ is right nilpotent.

5.1 Identities

In keeping with the spirit of the paper, we establish a number of identities that will be required to proceed further.

- (3) In case of Novikov algebras, the Jacobi identity splits into

$$[x, y]z + [y, z]x + [z, x]y = 0$$

and

- (4) $x[y, z] + y[z, x] + z[x, y] = 0$

- (5) Identity (2) can be written as

$$x[y, z] = (x, z, y) - (x, y, z)$$

$$(6) \quad (z, x, y) = [xy, z] - [x, z]y$$

$$(7) \quad (xy, z, t) = (x, z, t)y$$

$$(8) \quad (x, yz, t) = (x, y, t)z$$

$$(9) \quad (x, y, zt) = (x, yz, t) - (xy, z, t) + x(y, z, t) + (x, y, z)t$$

$$(10) \quad (x, y, zt) = (x, y, t)z - (x, z, t)y + x(y, z, t) + (x, y, z)t$$

Proof. (3) $[x, y]z + [y, z]x + [z, x]y = (xy)z - (yx)z + (yz)x - (zy)x + (zx)y - (xz)y = 0$

(4) Using the Jacobi identity and (3).

(6) Using $[xy, z] - x[y, z] - [x, z]y = (x, y, z) - (x, z, y) + (z, x, y)$ and (5).

(7), (8) Using (1) and (2).

(10) Using (7), (8) and (9). □

Lemma 5.1. *In any Novikov algebra N , the space of associators (N, N, N) is an ideal of N .*

Proof. The space (N, N, N) is a right ideal by (7) or (8). Then, from (9) or (10) we can conclude that this space is also a left ideal. □

Lemma 5.2. *Any Novikov algebra over a field of characteristic $\neq 2$ satisfies the following identities:*

$$(11) \quad (a, b, x) = \frac{1}{2}([ax, b] - [a, bx])$$

$$(12) \quad [a, b]x = \frac{1}{2}([ax, b] + [a, bx])$$

$$(13) \quad x[a, b] = [[x, a], b] + [a, [x, b]] + \frac{1}{2}([ax, b] + [a, bx]).$$

Proof. Applying identity (2), we get

$$(a, b, x) = (ab)x - a(bx) = (ab)x - [a, bx] - (bx)a = (ab)x - [a, bx] - (ba)x = [a, b]x - [a, bx]$$

Then, we can say:

$$(b, a, x) = [b, a]x - [b, ax]$$

By (1) we have,

$$2(a, b, x) = [a, b]x - [a, bx] + [b, a]x - [b, ax]$$

and,

$$[a, b]x - [a, bx] = [b, a]x - [b, ax]$$

which gives us (11)

$$2(a, b, x) = [ax, b] - [a, bx]$$

and (12)

$$2[a, b]x = [ax, b] + [a, bx]$$

. Using the Jacobi identity and (12) we have (13),

$$x[a, b] = [x, [a, b]] + [a, b]x = [[x, a], b] + [a, [x, b]] + \frac{1}{2}([ax, b] + [a, bx]).$$

□

Corollary 5.2.1. *Let N be a Novikov algebra over a field of characteristic $\neq 2$. Then the following statements are true:*

(i) *If I and J are right ideals of N , then $[I, J]$ is a right ideal of N .*

(ii) *If I and J are ideals of N , then $[I, J]$ is an ideal of N .*

6 Lie solvable Novikov algebra

A Novikov algebra is called *Lie-metabelian* if it satisfies the identity

$$(14) \quad [[x, y], [z, t]] = 0.$$

Lemma 6.1. *Any Lie-metabelian Novikov algebra N over a field of characteristic $\neq 2$ satisfies the identity:*

$$(15) \quad ([x, y], [z, t], s) = 0.$$

Proof. Using (14) and (12) we have,

$$[[a, b]x, [y, z]] = 0$$

. Then the identities (14) and (2) imply

$$\begin{aligned} & ([y, z], [a, b], x) \\ &= [y, z][a, b]x - [y, z]([a, b]x) \\ &= [a, b][y, z]x - [y, z]([a, b]x) \\ &= [a, b]x[y, z] - [y, z]([a, b]x) \\ &= [[a, b]x, [y, z]] = 0 \end{aligned}$$

□

Corollary 6.1.1. *The ideals (N, N, N) and $[N, N]$ of a Novikov algebra N over a field K of characteristic $\neq 2$ are associative and commutative and $(N, N, N) \subseteq [N, N]$.*

Proof. Note that (N, N, N) is an ideal of N by Lemma 4.1 and $[N, N]$ is an ideal of N by Corollary 4.2.1. The identity (11) implies that $(N, N, N) \subseteq [N, N]$. The identities (14) and (15) imply that $[N, N]$ is an associative and commutative algebra. □

Lemma 6.2. *Any Lie-metabelian Novikov algebra N over a field K of characteristic $\neq 2$ satisfies the identities:*

$$(16) \quad (x, [y, z], t)[a, b] = 0$$

$$(17) \quad ([x, y][z, t], a, b) = 0$$

$$(18) \quad [x, y][z, t](a, b, c) = 0.$$

Proof. The identities (13) and (15) imply that

$$(x[a, b], [y, z], t) = 0$$

then by (7),

$$(x, [y, z], t)[a, b] = (x[a, b], [y, z], t) = 0$$

i.e., (16) holds. Using (7), (1) and (16), we have,

$$([x, y][z, t], a, b) = ([x, y], a, b)[z, t] = (a, [x, y], b)[z, t] = 0$$

i.e., (17) holds. By (9),

$$[x, y][z, t](a, b, c) = ([x, y][z, t], a, bc) - ([x, y][z, t], ab, c) + ([x, y][z, t]a, b, c) - ([x, y][z, t], a, b)c$$

Using (17), from this we get (18). $\square$

Lemma 6.3. *Let N be a Lie-metabelian Novikov algebra over a field of characteristic $\neq 2$. Then $(N, N, N)^3 = [N, N]^4 = 0$.*

Proof. The identity (18) implies that $(N, N, N)^2(N, N, N) = 0$ since $(N, N, N) \subseteq [N, N]$ by Corollary 5.1.1. Consequently, $(N, N, N)^3 = 0$ since (N, N, N) is associative.

Note that $N[N, N] \subseteq (N, N, N)$ by (5). Consequently, $[N, N]^2 \subseteq (N, N, N)$. Then (18) implies that $[N, N]^2[N, N]^2 = 0$. This gives $[N, N]^4 = 0$ since $[N, N]$ is an associative algebra. $\square$

Theorem 6.4. *Let N be a Lie-solvable Novikov algebra of characteristic $\neq 2$. Then the ideal $[N, N]$ is right nilpotent.*

Proof. Let N be a Lie-solvable Novikov algebra with such that $N^{(n)} = 0$. We proceed using induction on n . By Lemma 5.3, this is true for $n = 2$. Suppose $n \geq 3$. Then $[N, N]$ is a Lie-solvable Novikov algebra such that $[N, N]^{n-1} = 0$. By the induction hypothesis, $[[N, N], [N, N]]$ is a right nilpotent ideal of N . Note that $[N, N]^4 \subseteq [[N, N], [N, N]]$. Hence, $[N, N]$ is a solvable ideal of N . Now, every solvable Novikov algebra is right-nilpotent. Hence, $[N, N]$ is a right nilpotent ideal of N . $\square$

References

- [1] K. Erdmann and M. J. Wildon, *Introduction to Lie Algebras*. Springer London, 2009.
- [2] K. Tulenbaev, U. Umirbaev, and V. Zhelyabin, “On the lie-solvability of novikov algebras,” 2021.